11.4.2 Iodine

The significance of Iodine as an essential trace element lies in its role in thyroxine production. Iodine is a constituent of thyroxine, the active principle of the thyroid gland. The thyroid gland plays an important role in energy metabolism and in the growth of the body.

Functions

- It is essential for the production of thyroid hormone called thyroxin which is secreted by the thyroid gland.
- Thyroxin controls the basic metabolic rate in the body as it controls the metabolism of all nutrients.
- Thyroxin regulates the rate of oxidation within the cells.
- It stimulates the physical and mental growth.
- It regulates the functioning of nerve and muscle tissue.

Food Sources of Iodine:

Iodine is present only in small amounts in common foods, the quantity of iodine present depending on the iodine content of the soil. Iodised salt, sea salt, vegetables grown in the sea shore, garlic, onion, cheese and sea fish are good sources of iodine.

Symptoms of Iodine deficiency:

- Wide variety of physical and neurological disorders associated with iodine deficiency are called **"Iodine Deficiency Disorders - IDD".**
- **Goitre:** It is characterized by swelling of thyroid gland.
- **Cretinism:** Person is deaf and has a shuffling gait, retarded mental and



Garlic



Salt



Seafish

Iodine Rich Foods



Cheese



Onion

Fig 11.22: Food Sources of Iodine

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